

Nonlocal reduced integrable mKdV-type equations from a vector integrable hierarchy

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This paper aims to present two hierarchies of nonlocal reduced integrable mKdV-type equations from a vector integrable hierarchy associated with a matrix Lie algebra, not being A type. The key point is to make similarity transformations for the spectral matrix, which keep the associated zero curvature equations invariant and then there follow reduced nonlocal integrable mKdV-type equations. The success lies in determining a Laurent series solution to the corresponding reduced stationary zero curvature equation, which generates temporal matrix spectral problems in the zero curvature formation.

Keywords: Matrix spectral problem; zero curvature equation; integrable hierarchy; nonlocal integrable equation; mKdV equations.

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1. Introduction

Integrable equations are associated with matrix spectral problems, from which diverse solutions can be presented by the inverse scattering transform, the Darboux transformation and the Riemann–Hilbert technique.^{1,2} A crucial step is to formulate a spectral matrix by determining a pseudoregular element in a matrix loop

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algebra.^{3,4} It is known that various integrable hierarchies are generated from the special linear algebras,^{1,5,6} and the special orthogonal algebras (see, e.g. Refs. 7 and 8). Recursion operators⁹ come naturally. Hamiltonian structures are furnished by the trace identity⁴ and the variational identity,¹⁰ depending on whether the associated matrix Lie algebra is semisimple or not. Combined with recursion structures that integrable hierarchies possess, bi-Hamiltonian structures,¹¹ which guarantee the Liouville integrability, are consequences of combining recursion operators and Hamiltonian structures.

Recently, similarity transformations are used to construct integrable reductions from known integrable hierarchies and the basic requirement is to keep the original zero curvature equations invariant. Various local and nonlocal reduced integrable nonlinear Schrödinger (NLS) and modified Korteweg-de Vries (mKdV) equations have been obtained this way from the Ablowitz–Kaup–Newell–Segur (AKNS) spectral problems (see, Refs. 12–17 and 18–23 for local and nonlocal reductions, respectively). Through pairs of similarity transformations, we can generate other kinds of reduced integrable equations, both local (see, Ref. 24) and nonlocal (see, e.g. Refs. 25–28). The Riemann–Hilbert technique has been used to solve reduced integrable equations, but there are still many challenging problems on nonlocal integrable equations.

The aim of this paper is to present an application of similarity transformations to a vector integrable hierarchy, to generate reduced nonlocal integrable mKdV-type equations. In Sec. 2, we recall a vector integrable hierarchy and its Abelian symmetry and conserved functional algebras. In Sec. 3, we propose two classes of similarity transformations for the involved spectral matrix, and work out two reduced integrable hierarchies of KdV-type equations. Two typical examples of nonlocal reduced integrable mKdV equations in the presented two reduced hierarchies are

$$\begin{aligned} r_t = & r_{xxx} - 3r_x\gamma\delta r^T(-x, -t)r - 3r\gamma\delta r^T(-x, -t)r_x \\ & + r_x\gamma r^T r(-x, -t)\delta + 2r\gamma r_x^T r(-x, -t)\delta, \end{aligned}$$

where γ and δ are two constant commuting symmetric and orthogonal matrices and r^T denotes the matrix transpose of the row vector potential r , and

$$\begin{aligned} r_t = & r_{xxx} + 3r_x\gamma\delta r^\dagger(-x, -t)r + 3r\gamma\delta r^\dagger(-x, -t)r_x \\ & - r_x\gamma r^T r^*(-x, -t)\delta - 2r\gamma r_x^T r^*(-x, -t)\delta, \end{aligned}$$

where γ and δ are two real commuting symmetric and orthogonal matrices, and $r^\dagger$ and r^* denote the Hermitian transpose and the complex conjugate of the row vector potential r , respectively. Here, γ and δ being two commuting symmetric and orthogonal matrices means that

$$\gamma\delta = \delta\gamma, \quad \gamma^T = \gamma, \quad \delta^T = \delta, \quad \gamma^T = \gamma^{-1}, \quad \delta^T = \delta^{-1}.$$

In the last section, we give a conclusion and some concluding remarks.

2. A Revisit of a Vector Integrable Hierarchy

Let us revisit a vector integrable Hamiltonian hierarchy, recently presented in Ref. 29, for the sake of subsequent analysis. The integrable hierarchy was generated from a matrix spectral problem associated with a matrix Lie algebra, not being A type.²⁹

Let $n \in \mathbb{N}$ be a given number and λ denote the spectral parameter (i.e. the eigenvalue parameter). We take a symmetric and orthogonal matrix γ of order n . Assume that the column vector potential u reads

$$u = u(x, t) = (r, s^T)^T, \quad r = r(x, t) = (r_1, \dots, r_n), \quad s = s(x, t) = (s_1, \dots, s_n)^T. \quad (2.1)$$

Beginning with the matrix spectral problem:

$$-i\phi_x = M\phi = M(u, \lambda)\phi, \quad M = \begin{bmatrix} -\lambda & r & 0 \\ s & 0 & \gamma^T r^T \\ 0 & s^T \gamma^T & \lambda \end{bmatrix}, \quad (2.2)$$

which presents a counterpart of the AKNS spectral problem,¹ we solve the associated stationary zero curvature equation

$$-iZ_x = [M, Z], \quad (2.3)$$

and assume a Laurent series solution:

$$Z = \begin{bmatrix} -e & f & 0 \\ g & h & \gamma^T f^T \\ 0 & g^T \gamma^T & e \end{bmatrix} = \sum_{l \geq 0} \lambda^{-l} Z^{[l]}, \quad (2.4)$$

with $Z^{[l]}$'s being defined by

$$Z^{[l]} = \begin{bmatrix} -e^{[l]} & f^{[l]} & 0 \\ g^{[l]} & h^{[l]} & \gamma^T f^{[l]T} \\ 0 & g^{[l]T} \gamma^T & e^{[l]} \end{bmatrix}, \quad l \geq 0. \quad (2.5)$$

It is direct to see that the stationary zero curvature equation with this solution presents the recursion relation for the solution Z :

$$\begin{cases} e_x^{[0]} = 0, & f^{[0]} = 0, & g^{[0]} = 0, & h_x^{[0]} = 0, \\ f^{[l+1]} = i f_x^{[l]} + r h^{[l]} + e^{[l]} r, & g^{[l+1]} = -i g_x^{[l]} + h^{[l]} s + s e^{[l]}, \\ e_x^{[l+1]} = i(f^{[l+1]} s - r g^{[l+1]}) = -f_x^{[l]} s - r g_x^{[l]}, \\ h_x^{[l+1]} = i(s f^{[l+1]} - g^{[l+1]} r - \gamma^T f^{[l+1]T} s^T \gamma^T + \gamma^T r^T g^{[l+1]T} \gamma^T), \end{cases} \quad (2.6)$$

where $l \geq 0$. Upon taking

$$e^{[0]} = 1, \quad h^{[0]} = 0, \quad (2.7)$$

and choosing the constant of integration as zero,

$$e^{[l]}|_{u=0} = 0, \quad h^{[l]}|_{u=0} = 0, \quad l \geq 1 \quad (2.8)$$

a sequence of $\{e^{[l]}, f^{[l]}, g^{[l]}, h^{[l]} \mid l \geq 1\}$ is uniquely determined. For example, we have

$$\begin{cases} f^{[1]} = r, & g^{[1]} = s, & e^{[1]} = 0, & h^{[1]} = 0; \\ f^{[2]} = ir_x, & g^{[2]} = -is_x, & e^{[2]} = -rs, & h^{[2]} = -sr + \gamma^T r^T s^T \gamma^T; \\ f^{[3]} = -r_{xx} - 2rsr + r\gamma^T r^T s^T \gamma^T, \\ g^{[3]} = -s_{xx} - 2srs + \gamma^T r^T s^T \gamma^T s, \\ e^{[3]} = -i(r_x s - r s_x), \\ h^{[3]} = i[-(sr_x - s_x r) + \gamma^T (r_x^T s^T - r^T s_x^T) \gamma^T]; \\ f^{[4]} = -ir_{xxx} - 3ir_x sr - 3irsr_x + ir_x \gamma^T r^T s^T \gamma^T + 2ir\gamma^T r_x^T s^T \alpha^T, \\ g^{[4]} = is_{xxx} + 3is_x rs + 3isrs_x - 2i\gamma^T r^T s_x^T \gamma^T s - i\gamma^T r^T s^T \gamma^T s_x. \end{cases} \quad (2.9)$$

As usual, the temporal matrix spectral problems are taken as

$$-i\phi_t = N^{[k]}\phi = N^{[k]}(u, \lambda)\phi, \quad N^{[k]} = (\lambda^k Z)_+ = \sum_{l=0}^k \lambda^l Z^{[k-l]}, \quad k \geq 0. \quad (2.10)$$

Now, the compatibility conditions of the two matrix spectral problems in (2.2) and (2.10), which are the zero curvature equations:

$$N_t - N_x^{[k]} + i[M, N^{[k]}] = 0, \quad k \geq 0, \quad (2.11)$$

generate the vector integrable hierarchy:

$$u_t = X^{[k]} = (if^{[k+1]}, -ig^{[k+1]T})^T, \quad \text{i.e. } r_t = if^{[k+1]}, \quad s_t = -ig^{[k+1]}, \quad k \geq 0 \quad (2.12)$$

the second example of nonlinear integrable equations is the generalized vector mKdV equations

$$\begin{cases} r_t = r_{xxx} + 3r_x sr + 3rsr_x - r_x \gamma r^T s^T \gamma - 2r\gamma r_x^T s^T \gamma, \\ s_t = s_{xxx} + 3s_x rs + 3srs_x - 2\gamma r^T s_x^T \gamma s - \gamma r^T s^T \gamma s_x, \end{cases} \quad (2.13)$$

where γ is an arbitrary symmetric and orthogonal matrix.

An application of the trace identity gives the Hamiltonian structure for the integrable hierarchy (2.12):

$$u_t = X_k = J \frac{\delta \mathcal{H}^{[k]}}{\delta u}, \quad k \geq 0, \quad (2.14)$$

where J is a Hamiltonian operator given by

$$J = i \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}, \quad (2.15)$$

with I_n being the identity matrix of order n , and the Hamiltonian functional $\mathcal{H}^{[k]}$ is defined by

$$\mathcal{H}^{[l]} = - \int \frac{e^{[l+2]}}{l+1} dx, \quad l \geq 0, \quad (2.16)$$

with $e^{[l]}$ being determined earlier by the stationary zero curvature equation. The Hamiltonian structure transforms conserved quantities to symmetries.³⁰ Infinitely many symmetries commute with each other:

$$[[X_k, X_l]] = \frac{d}{d\varepsilon} [(X_k(u + \varepsilon X_l) - X_l(u + \varepsilon X_k))]|_{\varepsilon=0} = 0, \quad k, l \geq 0, \quad (2.17)$$

guaranteed by the corresponding Lax operator:

$$\begin{aligned} [[V^{[k]}, V^{[l]}]] &= \frac{d}{d\varepsilon} [(V^{[k]}(u + \varepsilon X_l) - V^{[l]}(u + \varepsilon X_k))]|_{\varepsilon=0} \\ &+ [V^{[k]}, V^{[l]}] = 0, \quad k, l \geq 0. \end{aligned} \quad (2.18)$$

Moreover, the associated Hamiltonian structure in (2.14) guarantees the Abelian conserved functional algebra:

$$\{\mathcal{H}^{[k]}, \mathcal{H}^{[l]}\}_J = \int \left(\frac{\delta \mathcal{H}^{[k]}}{\delta u} \right)^T J \frac{\delta \mathcal{H}^{[l]}}{\delta u} dx = 0, \quad k, l \geq 0. \quad (2.19)$$

3. Nonlocal Reduced Integrable mKdV-type Equations

We take another symmetric and orthogonal matrix δ of order n , which commutes with γ involved in the spectral matrix M by (2.2), and introduce a bigger orthogonal matrix Σ as follows:

$$\Sigma = \begin{bmatrix} 0 & 0 & 1 \\ 0 & \delta & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad (3.1)$$

from which we can have

$$\Sigma M(\lambda) \Sigma^{-1} = \begin{bmatrix} \lambda & s^T \gamma \delta & 0 \\ \gamma \delta r^T & 0 & \delta s \\ 0 & r \delta^T & -\lambda \end{bmatrix}. \quad (3.2)$$

3.1. First class of reduced integrable mKdV-type equations

First, we take a similarity transformation for the spectral matrix M :

$$\Sigma M(x, t, \lambda) \Sigma^{-1} = -M(-x, -t, \lambda), \quad (3.3)$$

where Σ is defined by (3.1) and

$$M(x, t, \lambda) = M(u(x, t), \lambda), \quad M(-x, -t, \lambda) = M(u(-x, -t), \lambda).$$

Upon observing (3.2), we see that the above transformation equivalently requires

$$s = -\gamma\delta r^T(-x, -t) \quad \text{or} \quad r = -s^T(-x, -t)\gamma\delta. \quad (3.4)$$

Moreover, there follows

$$\Sigma Z(x, t, \lambda)\Sigma^{-1} = -Z(-x, -t, \lambda), \quad (3.5)$$

since both Laurent series matrices $\Sigma Z(x, t\lambda)\Sigma^{-1}$ and $Z(-x, -t, \lambda)$ of λ solve the stationary zero curvature Eq. (2.3), with $M(x, t, \lambda)$ being replaced by $-M(-x, -t, \lambda)$, and take the opposite initial values at $\lambda = \infty$. Therefore, by virtue of $N^{[k]} = (\lambda^k Z)_+$, we can have

$$\Sigma N^{[k]}(x, t, \lambda)\Sigma^{-1} = -N^{[k]}(-x, -t, \lambda), \quad k \geq 0, \quad (3.6)$$

which generates a relation

$$\begin{aligned} & \Sigma(M_t(x, t, \lambda) - N_x^{[r]}(x, t, \lambda) + i[M(x, t\lambda), N^{[r]}(x, t, \lambda)])\Sigma^{-1} \\ &= \Sigma_t(-x, -t, \lambda) - N_x^{[k]}(-x, -t, \lambda) \\ &+ i[M(-x, -t, \lambda), N^{[k]}(-x, -t, \lambda)], \quad k \geq 0, \end{aligned} \quad (3.7)$$

between the transformed zero curvature equations and the untransformed zero curvature equations. Obviously, this entails a reduced hierarchy of nonlocal integrable mKdV-type equations:

$$r_t = if^{[k+1]}|_{s=-\gamma\delta s^T(-x, -t)}, \quad k \geq 0, \quad (3.8)$$

each of which possesses infinitely many symmetries and conserved functionals inherited from the original ones, under the potential reduction (3.4). The nonlocal reduced integrable mKdV equation reads

$$\begin{aligned} r_t &= r_{xxx} - 3r_x\gamma\delta r^T(-x, -t)r - 3r\gamma\delta r^T(-x, -t)r_x \\ &+ r_x\gamma r^T r(-x, -t)\delta + 2r\gamma r_x^T r(-x, -t)\delta, \end{aligned} \quad (3.9)$$

where γ and δ are two commuting symmetric and orthogonal matrices and r^T denotes the matrix transpose of the potential r .

3.2. Second class of reduced integrable mKdV-type equations

Second, we take another similarity transformation for the spectral matrix M :

$$\Sigma M(x, t, \lambda)\Sigma^{-1} = M^*(-x, -t, -\lambda^*), \quad (3.10)$$

where Σ is defined by (3.1), $*$ stands for the complex conjugate, and

$$M(x, t, \lambda) = M(u(x, t), \lambda), \quad M(-x, -t, -\lambda^*) = M(u(-x, -t), -\lambda^*).$$

Once recognizing (3.2), the above transformation on the spectral matrix exactly leads to

$$s(x, t) = \gamma\delta r^\dagger(-x, -t) \quad \text{or} \quad r(x, t) = s^\dagger(-x, -t)\gamma\delta, \quad (3.11)$$

where γ and δ are assumed to be real. Under such a potential reduction, we can have

$$\Sigma Z(x, t, \lambda) \Sigma^{-1} = -Z^*(-x, -t, -\lambda^*), \quad (3.12)$$

since both Laurent series matrices $\Sigma Z(x, t\lambda) \Sigma^{-1}$ and $Z^*(-x, t, -\lambda^*)$ of λ do solve the stationary zero curvature Eq. (2.3), with $M(x, t, \lambda)$ being replaced by $M^*(-x, -t, -\lambda^*)$, and possess the opposite initial values at $\lambda = \infty$. Then, there follows

$$\Sigma N^{[2l+1]}(x, t, \lambda) \Sigma^{-1} = N^{[2l+1]*}(-x, -t, -\lambda^*), \quad l \geq 0, \quad (3.13)$$

and thus, we have

$$\begin{aligned} & \Sigma(M_t(x, t, \lambda) - N_x^{[2l+1]}(x, t, \lambda) + i[M(x, t, \lambda), N^{[2l+1]}(x, t, \lambda)]) \Sigma^{-1} \\ &= -(M_t(-x, -t, -\lambda^*) - N_x^{[2l+1]}(-x, -t, -\lambda^*) \\ &+ i[M(-x, -t, -\lambda^*), N^{[2l+1]}(-x, -t, -\lambda^*)])^*, \quad l \geq 0. \end{aligned} \quad (3.14)$$

This further tells that we obtain a reduced hierarchy of nonlocal integrable mKdV-type equations:

$$r_t = i f^{[2k+2]}|_{s=\gamma\delta r^\dagger(-x, -t)}, \quad k \geq 0, \quad (3.15)$$

whose infinitely many symmetries and conserved functionals are similarly inherited from the original ones, under the potential reduction (3.11). The nonlocal reduced integrable mKdV equation in this hierarchy reads

$$\begin{aligned} r_t &= r_{xxx} + 3r_x \gamma \delta r^\dagger(-x, -t)r + 3r \gamma \delta r^\dagger(-x, -t)r_x \\ &- r_x \gamma r^T r^*(-x, -t)\delta - 2r \gamma r_x^T r^*(-x, -t)\delta, \end{aligned} \quad (3.16)$$

where γ and δ are two real commuting symmetric and orthogonal matrices, and $r^\dagger$ and r^* denote the Hermitian transpose and the complex conjugate of the potential r , respectively.

Those two reduced hierarchies of nonlocal integrable mKdV-type equations are different from the previous ones generated from the standard AKNS integrable hierarchies (see, e.g. Refs. 19 and 20).

4. Concluding Remarks

Two reduced hierarchies of nonlocal integrable mKdV-type equations were generated from the novel vector integrable hierarchy associated with a matrix Lie algebra, not being A type. A crucial step is to formulate two similarity transformations, which keep the corresponding reduced zero curvature equations invariant. The given vector integrable hierarchy was reduced to two new nonlocal integrable hierarchies, one of which is real and the other is complex, and each of which involves two coordinates (x, t) and $(-x, -t)$.

We remark that it would be very interesting to see if the same idea of using similarity transformations could be applied to more other matrix spectral problems (see, e.g. Ref. 31). Such studies work towards to classify nonlocal integrable equations, based on Lax pairs of matrix spectral problems. Extensions of the nonlocal integrability theory to fractional differential and integro-differential equations (see, e.g. Refs. 32–34) should be interesting, too. It would also be significantly important to formulate soliton solutions of the presented nonlocal integrable equations by the Darboux transformation, Riemann–Hilbert problems or the Hirota direct method. Other solutions are important as well, which include lump solutions,^{35–37} complexitons,³⁸ rogue waves,^{39,40} and algebro-geometric solutions.^{41,42} For the presented integrable mKdV equations in the reduced hierarchies, multi-phase solutions, including solitary wave solutions, could be computed by using special functions, particularly elliptic functions. Until now, very little has been known about applications of nonlocal integrable equations.

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